

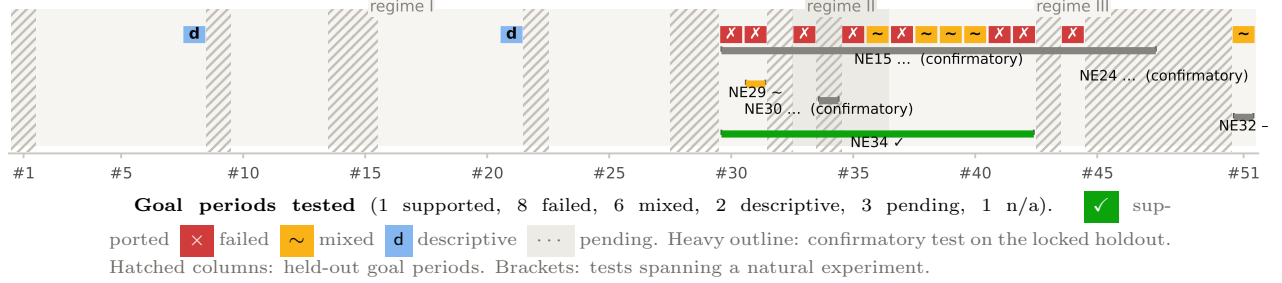
H01: Emergent superagents exist

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 2/5

Bottom line. No effective superagents: the persistent unit is an agent plus its own artifact, driven by the prompt, and grouping adds no detectable agency.

level: hypothesis · pipeline checks 84% · updated 2026-10-04



WHAT HAPPENED

Round 2 asked whether groups of agents plus their shared artifacts form *effective superagents*: units that use Kolchinsky–Wolpert (KW) semantic information to stay viable, with that information held at the group level. Tested on the work ledger (58k strict git/deploy writes) in 19 non-holdout units (#30–#51). Candidates were coordination-based (artifact crews, behavior-synchrony, co-allocation and reply communities), with rooms and labs as baselines. **No candidate is an individuality maximum** (crews’ Stouffer $Z = +0.06$; 2–8% local maxima). None carries KW semantic information about its environment ($\Delta V_{st} > 0$ in 2/19 units, ΔV_{obs} in 0/19), and none repairs itself beyond aggregation. Allocation does persist across nights ($\Delta C = 0.47$) and memory loss (16/17), but single agents out-persist crews. A goal change cuts advancement by 0.39. Holdout tests (NE30, NE24) written, **not run**.

PHYSICAL MODEL

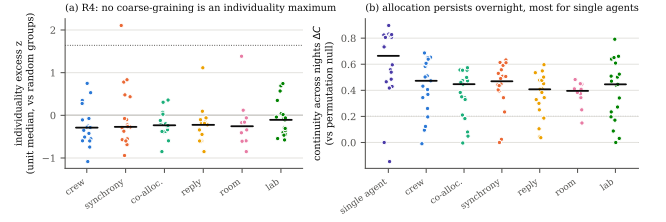
Model 04 (KW semantic information) on a coarse-grained unit $G = \text{members} + \text{shared artifacts } R_G + \text{channel}$. State x_b : did G advance R_G in 30-min bin b . Environment y_b : the rest of the swarm’s writes, and human or automated messages. V : advancement over horizon τ . p^f : KW’s coarse-graining scramble of $p(x_0, y_0)$. A^* : Krakauer et al.’s self-determined (organismal) predictive information; ι : its share of $H(x')$. ΔC : return to the unit’s artifacts after a night, versus a permutation null.

$$\Delta V_{st} = V[p] - V[p_X \otimes p_Y] = \mathbb{E}_p[(1 - e^{-\text{PMI}}) v(x_0, y_0)], \quad |\Delta V| \leq \sqrt{I/2}$$

$$S = \min_f \{I(p^f) : V[p^f] \approx V[p]\}, \quad \eta = S/I(X_0; Y_0)$$

$$\Delta V_{obs} : p(x_{t+1}|x_t, y_t) \rightarrow p(x_{t+1}|x_t)$$

$$A_G^* = I(x_{b+1}; x_b | y_b), \quad \iota_G = A_G^*/H(x_{b+1})$$



Observables. (a) Each dot is one unit’s median individuality excess over activity-matched random groups: every coarse-graining sits at the null. (b) Return to the unit’s artifacts the next morning, beyond a permutation null (membership excludes the scored day). Allocation persists everywhere, but most for single agents with their own projects. Rooms and labs are not special.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For someone running a swarm, the useful fact is a negative one: there is no group-level mind to steer. Coordination on artifacts adds no detectable self-maintenance beyond what the members already do. What persists is each agent’s attachment to its own artifacts, which survives every context erasure and memory loss, and the operator’s goal sets the work rate. The practical lever is therefore the artifact and the prompt, not the team. To keep work going across resets, make the work state legible in the repo; don’t count on memory or on group culture. Do teams matter? Not measurably here. The whole KW apparatus returns values near zero on this instrument, which is partly a limit of the instrument: the ledger counts attempts, not outcomes.

RESULTS VS PREDICTIONS

prediction	observed	verdict
R4 crews are max-ima	$Z = +0.06$; first in 5/18; 3.5% local max	failed [†]
R4d artifact units persist most	4/19; single agents top in 15/18	failed
R5a/b $\Delta V_{st}, \Delta V_{obs} > 0$	2/19 and 0/19	failed
R5c goal change costs viability	-0.39 ± 0.14 , $z = -2.75$	supported
R6b crews persist overnight	$\Delta C \geq 0.2$ in 15/19 (median 0.47)	supported
R6c survives memory loss	16/17 vs base 0.91	weak supp.
R6a/R8 spillover, repair	$Z = 2.4 < 3$; no repair	failed
R7 substrate independence	departures free; no turnover	mixed

[†]validated power $\leq 7\%$ at village sampling.

GOAL PERIODS IN DETAIL

NE15	...	pending (holdout)
NE24	...	pending (confirmatory, not run) (holdout)
NE30	...	pending (confirmatory, not run) (holdout)
G08	d	descriptive
G21	d	descriptive
G30	×	failed (round 2: no agency signature)
G31	×	failed (round 2: no agency signature)
G33	×	failed (round 2: no agency signature)
G35	×	failed (round 2: no agency signature; round 1: mixed)
G36	~	mixed (round 2: continuity, partial agency; round 1: failed)
G37	×	failed (round 2: no agency signature; round 1: failed)
G38	~	mixed (round 2: continuity, partial agency; round 1: mixed)
G39	~	mixed (round 2: continuity, partial agency; round 1: mixed)
G40	~	mixed (round 2: continuity, partial agency; round 1: supported)
G41	×	failed (round 2: no agency signature; round 1: mixed)
G42	×	failed (round 2: no agency signature; round 1: mixed)

+5 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	0	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

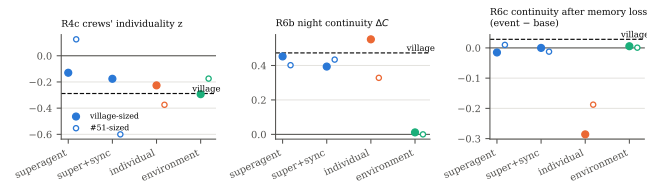
No replay. Goal changes bundle a new instruction with the scramble, and the only artifact-store scramble (NE24) is held out. The KW values rest on a coarse Markov kernel whose bootstrap is anti-conservative under hidden state, and the binary state can't see *which* artifact a unit works on. The ledger counts attempts (no failures, value or non-git work). Memory-loss evidence is 17 events, mostly in #51, where private goals can carry continuity.

ROUND 2

Run `confirm_r2.py`. H58: a coordination-first search with a which-artifact state. KW with outcome and context ledgers.

- **Essence.** effective superagents in Kolchinsky–Wolpert terms, running on a substrate of agents.
- **H01-R4** Individuality maxima pick out candidate superagents.
- **H01-R5** Superagents have positive group-level semantic information.
- **H01-R6** The agency is effective: it lives above the substrate.
- **H01-R7** Substrate independence.
- **H01-R8** The superagent does self-maintenance.

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Second figure. Synthetic worlds at village sampling (filled) and #51 sampling (open); the dashed line is the village. Timing individuality cannot tell the worlds apart. Night continuity rules out the purely environment-driven world. Continuity after memory loss rules out memory-held plans, and the village sits with the artifact-held worlds.